

IEORE 4004 - Quiz 1

11:45am - 12:55pm
Nov 11, 2024

This exam is composed of 10 pages and contains 2 exercises, plus a bonus exercise. The score for the bonus exercise will be computed separately and added as a bonus after the grades have been curved.

When solving an exercise, you can assume as known basic linear algebra facts, and all the statements seen in class and in the assignments, as well as closely related statements from the book and lecture notes. You cannot build on other results. If you have a doubt on whether you can use some result, ask us.

You cannot talk to your fellow students, for any reason. If you have a question, raise your hand and we will answer it. You are allowed to use one regular-sized cheatsheet, handwritten by yourself. Other than that, you are not allowed to use internet, books, notes,...

Motivate appropriately all your answers. There are many empty pages to use to write your solution, or as scratch paper. If you use all of them, you can request additional scratch paper. Submit everything, including the scratch paper and the cheatsheet.

Please do NOT open the exam until you are told to do so.

When you are done reading this page, write your name and uni below, then raise your hand.

Name: _____

UNI: _____

Good luck!

1. [50 points] Your company sells phones and owns one warehouse and one shop. In days 1,2,3, the shop needs to satisfy a given demand $d_1, d_2, d_3 > 0$ of phones, respectively. For $j \in \{1, 2, 3\}$, phones can be shipped from the warehouse to the shop in day j at a given per-unit cost $c_j > 0$. For $j \in \{1, 2\}$, phones already at the shop can be stored between day j and $j + 1$ at a given per-unit cost $q_j > 0$. Assume that the warehouse has enough capacity as to satisfy the demands of the shop.

1. Formulate the problem of shipping the phones from the warehouse to the shop in the different days as to satisfy the demand at minimum cost as an integer program.
2. Give an integer programming formulation for the following modification of the problem from part 1.: in day 2, the storing cost $q_2(x)$ is piecewise linear: in particular, the per-unit storing cost is given as follows

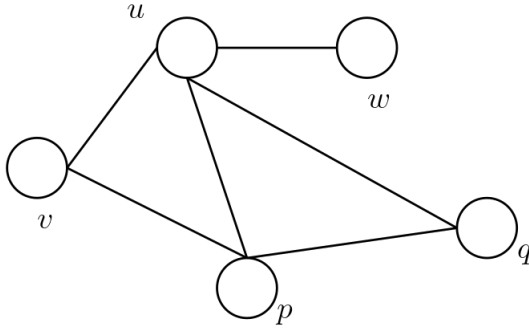
$$q_2 = \begin{cases} 10 & \text{if \# stored units} \leq 10 \\ 8 & \text{for every stored unit between 11 and 14} \\ 5 & \text{for every additional stored unit} \end{cases}$$

3. Give an integer programming formulation for the following modification of the problem from part 1.: We can decide to rent additional storage space in day 1. If we do so, we pay a fixed cost of k , but the per-unit storing cost q_1 is divided by 3. Else, we pay for storing costs as in part 1.

2. [50 points] Recall that a *dominating set* of a graph $G(V, E)$ is a set $S \subseteq V$ such that, for each vertex $i \in V$, there exists $j \in S$ such that either $i = j$, or $ij \in E$. The Minimum Dominating Set problem is the following:

- Given: A graph $G(V, E)$;
- Find: A dominating set of G of minimum cardinality.

1. Formulate the Minimum Dominating Set problem for the graph below as an integer program by explicitly listing the objective function and all the constraints.



2. Formulate the Minimum Dominating Set problem for a generic graph as an integer program.
3. Recall that a *vertex cover* of a graph $G(V, E)$ is a set of nodes $S \subseteq V$ such that, for each edge $uv \in E$, $\{u, v\} \cap S \neq \emptyset$. That is, for every edge of the graph, S contains at least one of the endpoints of the edge. Explain why a vertex cover of a graph $G(V, E)$ where each node is adjacent to at least one other node is also a dominating set of $G(V, E)$.

Bonus question. Let (M^*, p^*) be an equilibrium to an instance I of the assignment game. Let $\overline{M}$ be a matching of maximum weight ($\overline{M} \neq M^*$) in the usual graph where nodes are agents, and the weight of an edge (i, j) (where i is a buyer and j is a seller) is the valuation of buyer i for the house of seller j . Is $(\overline{M}, p^*)$ always an equilibrium for I ?